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# 1 Relative Strength Index

Relative strength is the ratio of average upward price movement to average downward price movement over a selected period:

$$RS = \frac{\text{Average gain}}{\text{Average loss}}.$$

Suppose the relative strength is

$$RS = 2.0.$$

This means that over the measured period, upward price movement has been twice as large as downward price movement.

- $RS > 1$: gains have been larger than losses.
- $RS = 1$: gains and losses have been equal.
- $RS < 1$: losses have been larger than gains.
- $RS \to \infty$: there have been gains but essentially no losses.
- $RS = 0$: there have been losses but no gains.

The Relative Strength Index, or RSI, converts relative strength into a bounded value between 0 and 100:

$$RSI = 100 - \frac{100}{1+RS} = 100\left(\frac{RS}{1+RS}\right).$$

As $RS \to \infty$, $RS/(1+RS) \to 1$, so $RSI \to 100$. When $RS = 1$, $RSI = 50$. When $RS \to 0$, $RSI \to 0$.

| **Relative Strength** | **RSI** |
|---|---|
| 0 | 0 |
| 0.25 | 20 |
| 0.5 | 33.33 |
| 1 | 50 |
| 2 | 66.67 |
| 3 | 75 |
| 4 | 80 |
| Very large | Approaches 100 |

RSI is commonly interpreted as follows:

- **Above 70:** frequently described as overbought.
- **Below 30:** frequently described as oversold.
- **Above 50:** upward momentum is dominant.
- **Below 50:** downward momentum is dominant.

To calculate the RSI for a time $t$, we need

$$RS_t = \frac{\text{AvgGain}_t}{\text{AvgLoss}_t}.$$

For each candlestick $c_j$, compute the close-to-close change.

$$\Delta_{c_j} = \text{Close}_{c_j} - \text{Close}_{c_{j-1}}.$$

The change is called a gain if it is greater than zero, it is called a loss if it is less than zero.

$$\text{Gain}_{c_j} = \max(\Delta_{c_j}, 0), \quad \text{Loss}_{c_j} = \max(-\Delta_{c_j}, 0).$$

Then, compute (for a 14-period RSI)

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14}.$$

## 1.1 Wilder's smoothing

Wilder's smoothing is a recursive moving-average method used in RSI, ATR, and several other indicators. It gives more weight to recent observations while retaining information from all earlier observations.

For a period $n$, the update formula is

$$S_t = \frac{(n-1)S_{t-1} + X_t}{n}.$$

- $X_t$ is the newest observation.
- $S_{t-1}$ is the previous smoothed value.
- $S_t$ is the new smoothed value.
- $n$ is the smoothing period.

For the relative strength, instead of using

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14},$$

we use Wilder's smoothing:

$$\begin{aligned} \text{AvgGain}_{c_j} &= \frac{13\text{AvgGain}_{c_{j-1}} + \text{Gain}_{c_j}}{14} \\ \text{AvgLoss}_{c_j} &= \frac{13\text{AvgLoss}_{c_{j-1}} + \text{Loss}_{c_j}}{14}. \end{aligned}$$

Then,

$$RS_{c_j} = \frac{\text{AvgGain}_{c_j}}{\text{AvgLoss}_{c_j}}.$$

Consequently,

$$RSI_{c_j} = 100 \left( \frac{RS_{c_j}}{1 + RS_{c_j}} \right).$$

The recursive formula requires a previous smoothed value, so the first value is normally initialized with a simple average of the first $n$ observations:

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

For the RSI, the initial average gain and initial average loss are simply the averages

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14}.$$

Every later value uses Wilder's recursive update.

## 2 Moving averages

A moving average is commonly used with time series data to smooth out short-term fluctuations and highlight longer-term trends or cycles

Given a series of numbers and a fixed subset size, the first element of the moving average is obtained by taking the average of the initial fixed subset of the number series. Then the subset is modified by ”shifting forward”; that is, excluding the first number of the series and including the next value in the series

A moving average is commonly used with time series data to smooth out short-term fluctuations and highlight longer-term trends or cycles - in this case the calculation is sometimes called a time average.

### 2.1 Simple Moving Average

In financial applications a simple moving average (SMA) is the unweighted mean of the previous $k$

$$\mathrm{SMA}_k = \frac{1}{k} \sum_{i=n-k+1}^{n} p_i.$$

When calculating the next mean $\mathrm{SMA}_{k,\mathrm{next}}$ with the same sampling width $k$ the range from $n-k+2$ to $n+1$ is considered. A new value $p_{n+1}$ comes into the sum and tho older value $p_{n-k+1}$ drops out. This simplifies the calculations by reusing the previous mean $\mathrm{SMA}_{k,\mathrm{prev}}$.

$$\begin{aligned}
\mathrm{SMA}_{k,\mathrm{next}} &= \frac{1}{k} \sum_{i=n-k+2}^{n+1} p_i \\
&= \frac{1}{k} \left( p_{n-k+2} + p_{n-k+3} + \cdots + p_n + p_{n+1} + \underbrace{p_{n-k+1} - p_{n-k+1}}_{0} \right) \\
&= \frac{1}{k} \left( p_{n-k+1} + p_{n-k+2} + \cdots + p_n \right) - \frac{p_{n-k+1}}{k} + \frac{p_{n+1}}{k} \\
&= \mathrm{SMA}_{k,\mathrm{prev}} + \frac{1}{k} \left( p_{n+1} - p_{n-k+1} \right).
\end{aligned}$$

### 2.2 Weighted Moving Average

A weighted average is an average that has multiplying factors to give different weights to data at different positions in the sample window.

In the financial field, and more specifically in the analyses of financial data, a weighted moving average (WMA) has the specific meaning of weights that decrease in arithmetical progression.[5] In an n-day WMA the latest day has weight n, the second latest $n-1$, etc., down to one.

$$\text{WMA}_M = \frac{np_M + (n-1)p_{M-1} + \cdots + 2p_{(M-n)+2} + p_{(M-n)+1}}{n + (n-1) + \cdots + 2 + 1}.$$

### 2.3 Exponential Moving Average

Exponential moving average (EMA) computes the weighted mean of all the previous data points and the weights decay exponentially. Concretely, given a sequence of data points

$$p_1, p_2, \cdots, p_t,$$

the EMA at data point $t$, denoted as $\mathrm{EMA}_t$ can be calculated with

$$\mathrm{EMA}_t = \begin{cases} p_1 & \text{if } t = 1 \\ \alpha p_t + (1-\alpha)\mathrm{EMA}_{t-1} & \text{otherwise} \end{cases}$$

where $\alpha \in (0, 1)$. Consider $t = 4$. Then,

$$\mathrm{EMA}_4 = \alpha p_4 + (1-\alpha)\left(\alpha p_3 + (1-\alpha)\left(p_2 + (1-\alpha)p_1\right)\right).$$

Expanding yields

$$\begin{aligned} \mathrm{EMA}_4 &= \alpha p_4 + (1-\alpha)\left(\alpha p_3 + \alpha(1-\alpha)p_2 + (1-\alpha)^2 p_1\right) \\ &= \alpha p_4 + \alpha(1-\alpha)p_3 + \alpha(1-\alpha)^2 p_2 + (1-\alpha)^3 p_1 \\ &= \alpha\left[p_4 + (1-\alpha)p_3 + (1-\alpha)^2 p_2\right] + (1-\alpha)^3 p_1. \end{aligned}$$

This should be convincing enough that in general, we have

$$\begin{aligned} \mathrm{EMA}_t &= \alpha \sum_{i=2}^{t} (1-\alpha)^{t-i} p_i + (1-\alpha)p_1 \\ &\approx \alpha \sum_{i=1}^{t} (1-\alpha)^{t-i} p_i. \end{aligned}$$

Notice that since $0 < \alpha < 1$, we indeed have exponential decay. Let's now consider the coefficients of the data points

$$\sum_{i=1}^{t} (1-\alpha)^{t-i}.$$

We can instead reverse the order of the sum, and say that

$$\sum_{i=1}^{t} (1-\alpha)^{t-i} = \sum_{i=1}^{t} (1-\alpha)^{i-1}.$$

Now, you may recognize this as a geometric series, since $|1-\alpha| < 1$. Recall that

$$\sum_{i=1}^{n} ar^{i-1} = \frac{a(1-r^n)}{1-r}, \quad r \neq 1.$$

Thus,

$$\sum_{i=1}^{t}(1-\alpha)^{i-1} = \frac{1-(1-\alpha)^t}{1-(1-\alpha)} = \frac{1-(1-\alpha)^t}{\alpha}.$$

As $t \to \infty$,

$$\frac{1-(1-\alpha)^t}{\alpha} \to \frac{1}{a}.$$

Since

$$a = \frac{1}{\frac{1}{\alpha}},$$

We can say that

$$\alpha\sum_{i=1}^{t}(1-\alpha)^{t-i}p_i = \frac{1}{\frac{1}{\alpha}}\sum_{i=1}^{t}(1-\alpha)^{t-i}p_i.$$

Hence, we have a weighted average whose weights are exponentially decayed.

Alpha ($\alpha$) is chosen either by using a standard time-period formula or by optimizing it for specific data stability and responsiveness

In finance and technical analysis, alpha is often derived directly from a chosen time window length (N) using the formula:

$$\alpha = \frac{2}{N+1}.$$

# 3 VWAP (Volume Weighted Average Price)

The average price a stock has traded at throughout the day weighted by volume. Instead of treating each price the same, VWAP assigns more weight to prices where trading activity was heavier.

# 4 Bollinger Bands

Bollinger Bands are a type of statistical chart characterizing the prices and volatility over time.

Bollinger Bands consist of an $N$-period moving average (MA), an upper band at $K$ times an $N$-period standard deviation above the moving average (MA + $K\sigma$), and a lower band at $K$ times and $N$-period standard deviation below the moving average (MA - $K\sigma$).

$N$ and $K$ are input parameters chosen independently by the user. Typical values for $N$ and $K$ are 20 days and 2, respectively. The default choice for the moving average is a simple moving average. Exponential is a common second choice.


## 4.1 Purpose

Bollinger bands gives us two types of signals, i.e. volatility and buy/sell.

The purpose of Bollinger Bands is to provide a relative definition of high and low prices of a market. By definition, prices are high at the upper band and low at the lower band.

The bands widen when a stock's price becomes more volatile and contract when it is more stable. Many traders see stocks as overbought as their price nears the upper band and oversold as they approach the lower band, signaling an opportune time to trade.

Bollinger Bands are a secondary indicator that is best used to confirm other analysis methods such as RSI and moving average convergence divergence (MACD). In particular, the use of oscillator-like Bollinger Bands will often be couped with a non-oscillator indicator-like chart pattern or a trendline. If these indicators confirm the recommendation of the Bollinger Bands, the trader will have great conviction that the bands are predicting correct price action in relation to market volatility.

## 4.2 Indicators derived from Bollinger Bands

### 4.2.1 Percent b

%b (pronoused "percent b") is derived from the forumla for **stochastics** 5 and shows where price is in relation to the bands. %b equals 1 at the upper band and 0 at the lower band. Writing *upperBB* for the upper Bollinger Band, *lowerBB* for the lower Bollinger Band, and **last** for the last (price) value:

$$\%b = (\mathrm{last} - \mathrm{lowerBB})/(\mathrm{upperBB} - \mathrm{lowerBB})$$

### 4.2.2 Bandwidth

*Bandwidth* tells how wide the Bollinger Bands are on a normalized basis. Writing the same symbols as before, and *middleBB* for the moving average:

$$\mathrm{Bandwidth} = (\mathrm{upperBB} - \mathrm{lowerBB})/\mathrm{middleBB}$$

# 5 Stochastics